Exam 3 Study Guide

Module 8: Pregnancy Complications · Module 9: Sudden Pregnancy Complications · Module 10: Complications of Labor & Birth / Surgical Interventions

Scoped to the Exam 3 Module Content Review (MCR): every topic it lists, built only from your uploaded materials (Chapter 20, 21, 23 and 24 slides, the weekly summaries, and the six lecture recordings). Pictures come from the course slides. The professor's emphasis comes from the recorded lectures (automated transcripts, so a few spots had audio gaps). Practice questions follow each module, with answers and rationales on the next pages; the dashed boxes are for your own notes.

THE 6 FUNCTIONS OF CLINICAL JUDGMENT (NCJMM), FROM YOUR MCR

1. Recognize cues — What matters most?

2. Analyze cues — What does it mean?

3. Prioritize hypotheses — Where do I start?

4. Generate solutions — What can I do?

5. Take action — What will I do?

6. Evaluate outcomes — Did it help?

Walk every topic through these steps: what cue matters, what it means, where you start, what you can do, what you will do, and whether it worked.

COLOR KEY

- **Teal** = Module 8 (Chapter 20)
- **Crimson** = Module 9 (Chapter 21)
- **Blue-indigo** = Module 10 (Chapters 23 & 24)
- **Gold** = Professor emphasized it in lecture
- **Violet** = Priority / safety: first action, can't-miss
- **Blue** = Memory tip (the professor's own analogies)
- **Dashed box** = your notes

FROM THE MCR

Module 8 · Chapter 20 | Module 9 · Chapter 21 | Module 10 · Chapters 23 & 24

Dosage calculations are also on the MCR, but none of your uploaded materials cover them. The MCR says to use faculty, the learning center, ATI modules and tutors for that piece.

The professor's study rule for Chapter 20 applies to the whole exam: **study what is on the slides**, not every condition in the textbook. Charts are not for memorizing, but you must be able to apply them to a case.

★ What the professor flagged + MCR checklist

START HERE

Paraphrased from the six lecture recordings. Right column = where it is in this guide.

Module 8 · pregnancy complications

Know your normals	Establish baselines; never dismiss a finding with "she's pregnant." Knowing normal pregnancy changes is how you eliminate wrong answers.	8.0
Left vs right heart failure	Lean on med-surg: right backs up into the body, left backs up into the lungs. Heart failure medications are not tested.	8.1
Positioning	No one in labor is supine. Orthopnea: sit her up first. Cardiac clients: head up, left tilt.	8.1
Anticoagulants	They do not bust the clot; the nursing concern is bleeding. Prevention matters most.	8.3
Asymptomatic UTI	Can climb to pyelonephritis, sepsis and preterm labor. Pyelo pain is reproduced by tapping over the kidney.	8.4
Diabetes	Insulin does not cross the placenta. Newborn hypoglycemia: feed within 30 minutes, heel stick. No need to memorize the GTT chart.	8.5

Module 9 · sudden complications

Any bleeding	Not normal in pregnancy; always investigate. A client who thinks she is miscarrying comes in to be examined.	9.0
Miscarriage types	Sort them by how much tissue has been lost. Missed abortion is linked to DIC.	9.1
Previa vs abruption	Painless bright red vs painful dark red with a board-like abdomen. No vaginal exams with previa.	9.5
DIC labs	Watch platelets and fibrinogen; D-dimer is raised in pregnancy anyway.	9.6
Terbutaline + betamethasone	Terbutaline is a "band-aid" that buys time for betamethasone. Hold terbutaline if HR is over 120. Betamethasone before 34 weeks.	9.7
Magnesium sulfate	Frequent assessment (DTRs, RR, LOC, urine). Toxicity: stop the infusion first, then level, calcium gluconate, provider.	9.8
Severe-range BP	Systolic over 160 or diastolic over 110 = stroke risk; antihypertensive within about 30 minutes.	9.8

Module 10 · labor & birth, surgical interventions

Bladder	Keep it empty: a full bladder blocks descent and contraction, and raises hemorrhage risk.	10.2
Oxytocin	High-alert drug; "it is not a race." Too-frequent contractions or a bad strip: turn it off first.	10.5
Cord prolapse	Lift the presenting part off the cord and never let go; positioning is only a band-aid.	10.7
Shoulder dystocia	McRoberts, suprapubic (never fundal) pressure, time head-to-body, check the arm and clavicle.	10.8
Vasa previa	The bleeding is fetal blood. Know accreta vs increta vs percreta.	10.12
Amniotomy	Fetal heart tones before and after is the most important step.	10.13
Cesarean	Classical incision = no vaginal birth later.	10.16

MCR checklist: tick it off when you can teach it without notes

Module 8

☐ High-risk pregnancy 8.0	☐ Left/right heart failure 8.1	☐ Chronic hypertension 8.2
☐ Venous thromboembolism 8.3	☐ UTI / pyelonephritis 8.4	☐ Diabetes / GDM 8.5

Module 9

☐ Spontaneous abortion 9.1	☐ Ectopic pregnancy 9.2	☐ Trophoblastic disease 9.3
☐ Premature cervical dilation 9.4	☐ Placenta previa 9.5	☐ Placental abruption 9.5

☐ DIC	9.6	☐ Preterm labor	9.7	☐ Hypertensive disorders	9.8
☐ HELLP syndrome	9.8	☐ Amniotic fluid volume	9.9	☐ Isoimmunization / RhoGAM	9.10
☐ Multiple gestation	9.11	☐		☐	

Module 10

☐ Dysfunctional labor	10.2	☐ GBS	10.1	☐ Induction / augmentation	10.4
☐ Cervical ripening / Bishop	10.4	☐ Uterine rupture / inversion	10.6	☐ Amniotic fluid embolism	10.6
☐ Umbilical cord prolapse	10.7	☐ Posterior / breech	10.9	☐ Macrosomia	10.8
☐ Shoulder dystocia	10.8	☐ Inlet / outlet contraction	10.10	☐ External cephalic version	10.9
☐ Forceps / vacuum	10.11	☐ Velamentous insertion	10.12	☐ Vasa previa	10.12
☐ Accreta / increta / percreta	10.12	☐ Amniotomy	10.13	☐ Episiotomy	10.14
☐ Internal monitoring	10.15	☐ Cesarean birth	10.16	☐ Practice questions	8.6 · 9.12 · 10.17

MODULE 8 · CHAPTER 20

Pregnancy Complications

A family experiencing a pregnancy complication from a preexisting or newly acquired illness. Focus only on the conditions on the slides; the textbook chapter covers far more than the exam.

8.0 High-risk pregnancy: definition and assessment

ASSESSMENT | MCR: MODULE 8

DEFINITION	A concurrent disorder, pregnancy-related complication, or external factor that jeopardizes the health of the woman, the fetus, or both. Often more than one factor; it can be internal or external.
WHO MANAGES IT	Maternal-fetal medicine (MFM) specialists follow the client closely and recommend the delivery plan to the OB (they usually do not deliver). Expect increased surveillance.
EFFECT ON BABY	Most conditions affect the fetus in some way. Even when a condition does not physically reach the baby, the psychological toll on the mother matters, and high stress itself can harm the baby.
OBJECTIVE DATA	Measurable: vital signs, blood pressure, weight (every visit), fundal height, labs, fetal heart rate, fetal movement felt by the provider, edema.
SUBJECTIVE DATA	Reported: pain, nausea, heartburn, anxiety, shortness of breath, fatigue, appetite changes, feeling the baby move.
NORMAL BUT LOOKS ABNORMAL	Edema of the feet, nausea and vomiting, frequent urination, appetite changes, slightly higher heart rate and respiratory rate, breast tenderness.
ALWAYS ABNORMAL	High blood pressure (still a concern in pregnancy). Rapid weight gain, such as 4 lb in a week, is a red flag in anyone.

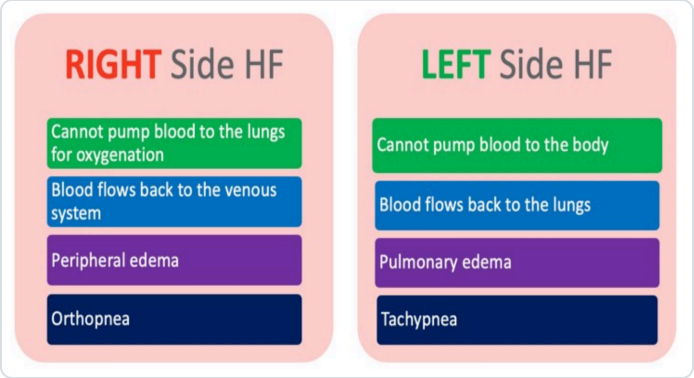
★ **PROFESSOR EMPHASIZED**

Establish each client's **baseline** so you can spot a significant change early. Never dismiss a finding with "she's pregnant, that's why" (example: a client who used to exercise is suddenly winded early in pregnancy, so investigate). **Treat the patient, not the monitor:** a client who reports contractions that do not show on the monitor is still a problem.

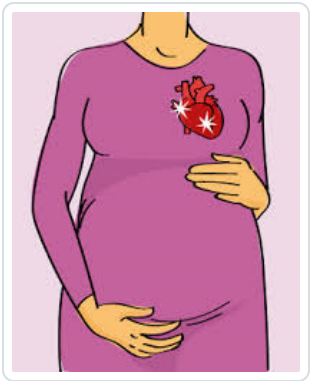
MY NOTES: HIGH-RISK PREGNANCY

8.1 Cardiovascular disease: left- vs right-sided heart failure

CARDIAC · MCR: MODULE 8



Right vs left side heart failure (Ch 20 slide)



Blood volume and cardiac output rise 30–50%

WHY PREGNANCY STRAINS THE HEART	Blood volume and cardiac output rise **30–50%**, gradually, peaking at **28–32 weeks** (when decompensation is most likely to show). Labor is the body's "super stress test."
BIGGEST CONCERN	The baby ("patient number two"): is the fetus getting enough oxygen, nutrients and blood flow if the mother cannot perfuse herself? **Left-sided failure is more detrimental to the fetus.**
PREEXISTING DISEASE	See MFM as soon as the first period is missed, so baselines and trends can be followed.

Type	What is happening	Signs and symptoms
Right-sided (backs up into the body)	Right ventricle cannot pump blood to the lungs; blood backs up into the venous system.	Peripheral edema, JVD, enlarged liver and spleen, ascites
Left-sided ("left = lungs")	Left ventricle cannot pump blood out to the body; blood backs up into the lungs.	Crackles, pulmonary edema, dyspnea, orthopnea, tachypnea, weak pulses, slow cap refill, fatigue

ORTHOPNEA	Hard to breathe lying down. First action: **sit her up** so fluid settles to the bottom of the lungs.
PAROXYSMAL NOCTURNAL DYSPNEA	Shortness of breath while sleeping: at rest, extra fluid moves back into circulation and worsens pulmonary edema.
ANTEPARTUM CARE	Promote rest (two rest periods a day plus a full night's sleep); avoid strenuous activity (may stop working early); low-sodium diet (avoid processed, packaged, canned foods; read labels); **daily weight at the same time**; take cardiac meds exactly as prescribed and report barriers such as cost; avoid infection (hand and dental hygiene; endocarditis risk).
INTRAPARTUM CARE	Frequent vital signs; continuous fetal monitoring; head of bed up and left tilt (never supine); **epidural** to reduce stress and bearing down; **assisted vaginal delivery** (forceps or vacuum) to reduce pushing.
POSTPARTUM CARE	Watch closely right after the placenta delivers: a massive shift of blood back into the systemic circulation (the uterus was getting about 500 mL/min) means high risk of cardiac collapse; hemorrhage risk is also highest then. Assess for heart failure; compression (SCD) boots if bed-bound; stool softeners; artificial valves may get prophylactic antibiotics.

★ **PROFESSOR EMPHASIZED**

Know right vs left failure cold; the pathophysiology is the same as med-surg. **Heart failure medications are not tested.** Intrapartum priorities (positioning, epidural, assisted delivery) all lower the heart's workload.

✎ **MEMORY TIP (FROM LECTURE)**

Left = lungs. Right side backs up into the body (edema, JVD). Pregnancy is "the body's natural stress test," and labor is the super stress test.

MY NOTES: CARDIAC DISEASE

8.2 Chronic hypertensive vascular disease

MCR: MODULE 8

On the MCR for Module 8. In lecture it was treated as both a vessel and a kidney problem: hypertensive disorders show up as high blood pressure plus poor kidney function, often with protein in the urine. The full detail (chronic and gestational hypertension, preeclampsia, eclampsia, HELLP, magnesium) is taught in Module 9: see **9.8**.

★ **PROFESSOR EMPHASIZED**

Protein in the urine at a routine prenatal dip prompts a preeclampsia work-up. Prenatal visits cannot always prevent these disorders, but they catch them early, which is why no visit should be skipped.

8.3 Venous thromboembolic disease (DVT)

HEMATOLOGIC | MCR: MODULE 8

VIRCHOW'S TRIAD	**Hypercoagulability, venous stasis, vessel injury.** Any clot fits one of these three, and pregnancy meets all three.
WHY PREGNANCY RAISES RISK	Hypercoagulable state (estrogen); blood pools in the legs and the uterus compresses the iliac veins and inferior vena cava; decreased peripheral vascular resistance; vessel injury during delivery or surgery.
OTHER RISK FACTORS	Immobility, estrogen birth control, long-distance travel, smoking and hypertension (vessel injury), obesity, surgery, hospitalization in the past 2 weeks, AFib or stroke.
CLINICAL MANIFESTATIONS	**Unilateral** edema, warmth, redness, pain and tenderness (worse with walking or standing). Measure and compare calf diameters.
DIAGNOSTICS	Ultrasound; PTT and INR help. **D-dimer is not reliable in pregnancy** (pregnancy alone raises it, as does surgery).
PREVENTION	Ambulation (best), hydration, SCD boots, prophylactic anticoagulant (e.g., Lovenox) for high-risk clients such as post-cesarean with BMI over 30 or prior DVT. Cesarean clients are usually up within about 8 hours.
TREATMENT	Anticoagulants thin the blood so the clot cannot grow and blood can flow around it; the body reabsorbs the clot. Delivery may be timed and the drug stopped about 24 hours before.
NURSING PRIORITY	Monitor for **bleeding** on anticoagulants.

★ **PROFESSOR EMPHASIZED**
Anticoagulants do **not** bust the clot. Prevention matters most because every pregnant client is already at risk.

MY NOTES: DVT

8.4 Renal and urinary disorders: UTI and pyelonephritis

RENAL · MCR: MODULE 8

CAUSES & RISK FACTORS	Ureters dilate (progesterone) and are compressed, so urine stagnates even when she voids often. About 10–15% of pregnant women get a UTI. Weekly summary causes: **asymptomatic bacteriuria, urinary stasis, glycosuria**. Usual organism: **E. coli** (ascending infection).
ROUTINE URINE DIP	Every prenatal visit: nitrites and WBCs (infection), glucose (diabetes clue, may prompt an early GTT), protein (preeclampsia work-up).
WHY ASYMPTOMATIC MATTERS	It can climb to the kidney (pyelonephritis), cause sepsis, and trigger preterm contractions and delivery.
UTI MANIFESTATIONS	Dysuria/burning, urgency, frequency, foul-smelling or cloudy concentrated urine, hematuria, possible fever.
PYELONEPHRITIS	Infection in the pelvis of the kidney. Severe flank pain that the provider reproduces by tapping over the kidney; fever; very sick. **Inpatient IV antibiotics**; expect uterine irritability and preterm contractions, so monitor the fetus; stay until fever-free and not contracting.
PYELO VS BACK LABOR	Labor pain is rhythmic and sweeps from the back around to the front. Pyelo pain sits over the kidney and hurts when tapped.
GBS IN URINE	If urine grows Group B strep (or the late third-trimester GBS culture is positive), she automatically gets antibiotics in labor (GBS is linked to newborn pneumonia). See 10.1.
CLIENT TEACHING	Wipe front to back; void completely and don't hold urine; void before and after intercourse; increase fluids; cotton underwear, loose clothes, no wet clothes; shower; no douching, bubble baths or bath bombs; cranberry juice (acidifies urine, bacteria can't stick to the bladder wall); finish prescribed antibiotics.

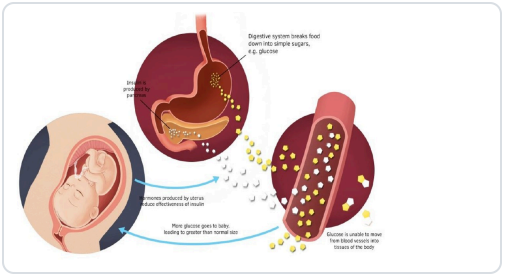
★ **PROFESSOR EMPHASIZED**
Why an asymptomatic UTI is dangerous (pyelo, sepsis, preterm labor), and how flank pain differs from back labor.

MY NOTES: UTI / PYELO

8.5 Diabetes in pregnancy and gestational diabetes (GDM)

ENDOCRINE | MCR: MODULE 8

TYPE 1	Autoimmune destruction of beta cells; no insulin; lifelong injections.
TYPE 2	Insulin resistance or too little insulin; diet, exercise, oral meds, sometimes insulin.
PREEXISTING DIABETES	Tight glucose control is paramount before pregnancy and in the **first trimester**, when organs form. High sugars raise the risk of anomalies, miscarriage and stillbirth.
PATHOPHYSIOLOGY (GDM)	Pregnancy hormones (hPL, estrogen, progesterone) make cells resistant to insulin, causing high glucose after meals.



Gestational diabetes (Ch 20 slide)

EFFECT ON BABY	**Insulin does not cross the placenta; glucose does.** The baby makes its own insulin from about 16–18 weeks, which acts like a growth hormone, so the baby grows large (macrosomia: big cheeks and face; shoulder dystocia risk).
NEWBORN AFTER BIRTH	The cord is cut, the sugar supply stops, but the baby's insulin is still high, so glucose **plummets**. Feed within 30 minutes; heel-stick glucose (side of the heel); watch for jitteriness, lethargy, poor feeding, crying. Very low sugars may need NICU.
WHO IS SCREENED EARLY	Previous GDM, BMI over 30, glucose in the urine, family history, a previous baby over 9 lb or a large fetus, previous unexplained fetal loss. (Type 1 and type 2 clients are not tested; they are already diagnosed.)
DIAGNOSTIC PROCEDURES	**Everyone at 24–28 weeks.** 1-hour test: 50 g glucose drink, non-fasting, draw at 1 hour; over 140 mg/dL means she needs the 3-hour test. 3-hour test: fasting draw, 100 g drink, then draws at 1, 2 and 3 hours (four draws). **Two abnormal values = GDM.**

3-hour GTT (slide, Table 20.3)	Fasting	1 hour	2 hours	3 hours
Threshold (mg/dL)	95	180	155	140

NUTRITION	Carb control: complex instead of simple carbs; fruit counts as carbs; dietitian; carbs about 40–50% of intake.
MONITORING	Fasting and after-meal finger sticks with a food and glucose diary. Ideal maternal glucose under 90.
EXERCISE	About 30 minutes daily; exercise makes cells more sensitive to insulin.
MEDICAL MANAGEMENT	Many control it with diet and exercise. Insulin needs rise as the baby grows (2nd and 3rd trimesters), so doses are adjusted.
SURVEILLANCE	Growth ultrasounds, possibly weekly non-stress tests, placental function tests, kick counts.
POSTPARTUM TEACHING	Glucose usually normalizes after delivery, but she is about 50% more likely to develop type 2 diabetes later.

★ **PROFESSOR EMPHASIZED**

Insulin does not cross the placenta. You do **not** need to memorize the GTT chart (values will be given), but know the **two-abnormal-values rule**. Her practice example had an abnormal fasting and 1-hour value, so it was a GDM diagnosis.

✎ **MEMORY TIP (FROM LECTURE)**

"Sugar mama": mom sends extra sugar, the baby makes extra insulin and grows. Cut the cord, the sugar mama is gone, and the insulin is still high, so the baby's sugar drops.

MY NOTES: DIABETES / GDM

8.6 Module 8 practice: check yourself

NGN-STYLE · ANSWERS ON THE NEXT PAGE

CASE
8 weeks pregnant with known cardiac disease, followed by MFM. Later in pregnancy she develops gestational diabetes. (Built from the lecture's cardiac NCLEX item and the GDM content.)

M8-1 · MC · recognize cues **Which statement by this client at 8 weeks needs follow-up?**

A. "I've been really constipated."
B. "I'm nauseated most mornings."
C. "I gained 4 pounds this week and my ankles are swollen."
D. "I have to urinate all the time."

M8-2 · SATA · analyze cues **Which findings fit left-sided heart failure? Select all that apply.**

A. Crackles in the lungs
B. Jugular vein distension
C. Orthopnea
D. Ascites
E. Weak peripheral pulses
F. Dyspnea

M8-3 · MC · take action **In labor, she is lying flat and struggling to breathe. What is the nurse's first action?**

A. Increase the IV rate
B. Raise the head of the bed and tilt her to the side
C. Call for an epidural
D. Apply a fetal scalp electrode

M8-4 · SATA · generate solutions **Which intrapartum interventions reduce cardiac workload for this client? Select all that apply.**

A. Epidural anesthesia
B. Supine position for monitoring
C. Assisted vaginal delivery (forceps or vacuum)
D. Prolonged pushing in second stage
E. Frequent vital signs
F. Continuous fetal monitoring

M8-5 · MC · prioritize **When is she at the highest risk of cardiac collapse?**

A. At 12 weeks
B. During early latent labor
C. Right after the placenta delivers
D. At the 6-week postpartum visit

M8-6 · Matrix · analyze cues **Her 3-hour GTT results are below (thresholds will be given: fasting 95, 1 h 180, 2 h 155, 3 h 140). Mark each value, then decide: does she have GDM?**

Finding	Normal	Abnormal
Fasting: 98 mg/dL	○	○
1 hour: 185 mg/dL	○	○
2 hours: 150 mg/dL	○	○
3 hours: 130 mg/dL	○	○

M8-7 · MC · take action **Her newborn (10 lb) is jittery and feeding poorly at 1 hour of life. What is the priority?**

A. Swaddle and recheck in 2 hours
B. Heel-stick blood glucose and feed
C. Give the first bath
D. Check the baby's temperature only

M8-8 · MC · analyze cues **Why does this newborn's blood sugar drop after birth?**

A. Maternal insulin crossed the placenta
B. The baby's own insulin is still high after the maternal glucose supply is cut off
C. The baby has type 1 diabetes
D. Formula is low in sugar

M8-9 · SATA · teaching **Which statements show she understands UTI prevention? Select all that apply.**

A. "I'll wipe front to back."
B. "I'll void before and after intercourse."
C. "Bubble baths will help me relax."
D. "I'll hold my urine until it's convenient."
E. "I'll drink more fluids and wear cotton underwear."

M8-10 · MC · analyze cues **At 30 weeks she has fever and pain that increases when the provider taps over her kidney. What does this suggest?**

A. Back labor
B. Pyelonephritis
C. Placental abruption
D. Round ligament pain

M8-11 · MC · recognize cues **After a cesarean, which finding most suggests a DVT?**

A. Bilateral ankle edema
B. One calf warm, red and larger than the other
C. D-dimer above the non-pregnant range
D. Pain at the incision

M8-12 · MC · evaluate **She is started on an anticoagulant. Which teaching statement is correct?**

A. "This medicine will dissolve the clot."
B. "It keeps the clot from getting bigger; report any unusual bleeding."
C. "I can stop it once my leg feels better."
D. "It works by raising my platelets."

8.7 Module 8 answer key & rationales

CHECK AFTER YOU TRY

M8-1 **C**
Rapid weight gain (4 lb in a week) with swelling is a red flag at any gestation, especially with cardiac disease. Constipation, nausea and frequency are expected early in pregnancy.

M8-2 **A, C, E, F**
Left = lungs: crackles, dyspnea, orthopnea, and weak pulses from poor output to the body. JVD and ascites are right-sided (backing up into the body).

M8-3 **B**
Orthopnea: sit her up so fluid settles; no laboring client should be supine. Side tilt helps perfusion.

M8-4 **A, C, E, F**
Epidural reduces stress and bearing down; assisted delivery reduces pushing; frequent VS and continuous monitoring are intrapartum care. Supine and prolonged pushing add workload.

M8-5 **C**
After the placenta delivers, a large volume of blood shifts back into the systemic circulation; the heart must adapt. Hemorrhage risk is also highest then.

M8-6 **Fasting and 1 hour abnormal → GDM: yes**
Two abnormal values out of four make the diagnosis (the professor's own example used an abnormal fasting and 1-hour value).

M8-7 **B**
Newborns of diabetic mothers are at risk for hypoglycemia (jittery, lethargic, poor feeding). Check glucose by heel stick and feed (within 30 minutes of birth).

M8-8 **B**
Insulin does not cross the placenta. The baby made extra insulin in response to maternal glucose; when the cord is cut the supply stops but insulin stays high.

M8-9 **A, B, E**
Bubble baths irritate the urethra; holding urine causes stasis.

M8-10 **B**
Pyelonephritis pain is over the kidney and reproduced by tapping, with fever. Back labor is rhythmic and sweeps from back to front.

M8-11 **B**
DVT is usually unilateral: warmth, redness, swelling, compare calf diameters. D-dimer is unreliable in pregnancy and after surgery.

M8-12 **B**
Anticoagulants do not bust the clot; they keep it from growing so blood can flow around it. The nursing concern is bleeding.

MODULE 9 · CHAPTER 21

Sudden Pregnancy Complications

Bleeding, preterm labor, hypertensive disorders, and the conditions that put a pregnant woman and her fetus at sudden risk.

9.0 Assessing vaginal bleeding

ASSESSMENT · MCR: MODULE 9

Any bleeding in pregnancy is not normal and is reported and investigated. It may be benign, but it can come from something that sustains the pregnancy.

CONFIRM PREGNANCY	Has she confirmed it? Early losses can happen before she knows. Separate a period from bleeding in pregnancy.
GESTATIONAL AGE	Needed before anything else makes sense.
ONSET & DURATION	How long, still happening, what was she doing (lifting, cleaning).
AMOUNT & QUALITY	Brown vs bright red vs dark red; fluid; clots; tissue. "Small amount" means different things to different people, so quantify (saved pads or towels, saved tissue).
PAIN	Any pain, and where.
BLOOD TYPE	Rh status (Rh-negative mothers need RhoGAM; see 9.10).
SIGNS OF HYPOVOLEMIC SHOCK	Falling BP, racing heart, fast breathing, low SpO2, pale, cool and clammy, diaphoretic, fatigue, confusion, anxiety or a sense of doom.

Trimester	Common causes (any can occur at any time)
First	Spontaneous abortion; ectopic pregnancy
Second	Gestational trophoblastic disease (molar pregnancy); premature cervical dilation
Third	Placenta previa; placental abruption; preterm labor

★ **PROFESSOR EMPHASIZED**

A client who thinks she is miscarrying should **come in to be examined and save any tissue**. Lecture NCLEX item: wrong answers were bed rest for 24 hours (won't stop it, gives no information), "spotting is harmless" (dismissive), and a tampon (absorbs blood you need to quantify; foreign body). Only the right answer was an assessment.

9.1 Spontaneous abortion (miscarriage)

1ST TRIMESTER · MCR: MODULE 9

Pregnancy loss before 20 weeks (medical term; not the same as elective termination, so choose words carefully). About 1 in 5 pregnancies. Usual cause: a development problem (often chromosomal) not compatible with life.

Type	Cervix / tissue	Fetal heart tones	Key cues and care
Threatened	Closed; no tissue passed	Present	Spotting or brown discharge, no pain, not contracting. Most hopeful; follow up with the OB.

Imminent (inevitable)	Dilated; membranes bulging; no tissue yet	Usually present	Contractions, dilation, bleeding. Cannot be stopped.
Incomplete	Delivery started; fetal parts felt; some products remain	Variable	More cramping and bleeding; started but not finished.
Complete	All products passed	Absent	Everything has passed (she may bring tissue in).
Missed → septic	No labor signs, no cervical change	Absent	She may not know. Tissue becomes necrotic and infected: septic shock. Still must be delivered.

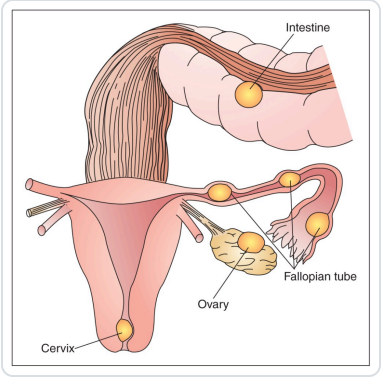
★ PROFESSOR EMPHASIZED
Sort the types by **how much tissue has been lost**, not by memorizing pain and bleeding descriptions. A missed abortion (like a retained stillbirth) is strongly linked to **DIC**. Bed rest and pressure on the cervix do not stop an imminent loss.

MY NOTES: MISCARRIAGE

9.2 Ectopic pregnancy

1ST TRIMESTER · EMERGENCY IF RUPTURED · MCR: MODULE 9

PATHOPHYSIOLOGY	The fertilized egg implants outside the uterus, usually in the fallopian tube. It gets a blood supply and grows, but it is not viable. Can happen in a first pregnancy.
CAUSES & RISK FACTORS	IUD in place; tubal ligation; **pelvic inflammatory disease (a "biggie")**; STIs; previous tubal or pelvic surgery; endometriosis; PCOS: anything that scars, narrows or damages the tube.
UNRUPTURED	Positive pregnancy test, fullness or bloating at the site, discomfort.



Ectopic implantation sites (Ch 21 slide)

RUPTURED (EMERGENCY)	Sudden severe one-sided lower abdominal pain plus hypovolemic shock from internal bleeding. Lecture case: BP 95/58, HR 132, RR 24, SpO2 92%, diaphoretic, pale.
MEDICAL MANAGEMENT	Unruptured: **methotrexate** (shrinks and eliminates the cells). Ruptured: **surgery** to stop the bleed; damaged tube removed (higher risk of another ectopic and infertility).
METHOTREXATE TEACHING	Stop prenatal vitamins and folic acid (they support cell growth); no alcohol (severe vomiting); serial hCG levels until they drop; wait a few months before conceiving so hCG is reliable.
RH	Rh-negative clients get RhoGAM.

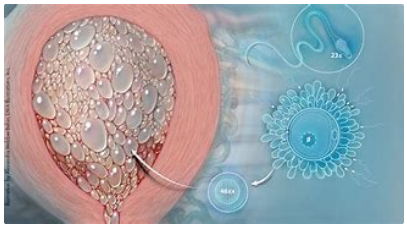
★ PROFESSOR EMPHASIZED
Don't just memorize the risk factors; they make sense (an IUD blocks implantation in the uterus; a ligation or scarring blocks the tube). **Left-sided pain + shock signs + a surprise pregnancy with an IUD = think ectopic.**

MY NOTES: ECTOPIC

9.3 Gestational trophoblastic disease (molar pregnancy)

2ND TRIMESTER | MCR: MODULE 9

PATHOPHYSIOLOGY	The cells that should become the placenta (trophoblasts) multiply abnormally and fill with fluid. No viable fetus; scrambled genetic material. Cannot be predicted or prevented.
RISK FACTORS	Clomid use, age over 40, more common in the Asian population.
CLINICAL MANIFESTATIONS	Very high, rising hCG; uterus larger than dates (case: 13 weeks, fundus near the umbilicus ≈ 20 weeks); severe morning sickness; dark brown "sludgy" bleeding with grape-like tissue; anemia; early preeclampsia signs.
DIAGNOSTICS	Ultrasound: dense cluster of vesicles, no fetus ("snowstorm").
MEDICAL MANAGEMENT	D&C with suction; methotrexate if tissue may remain.
FOLLOW-UP TEACHING	Serial hCG levels. Delay pregnancy 6 months to 1 year: a rising hCG could be a regrowing mole or a new pregnancy and you can't tell which. Tissue left behind can regrow: risk of metastasis (choriocarcinoma).



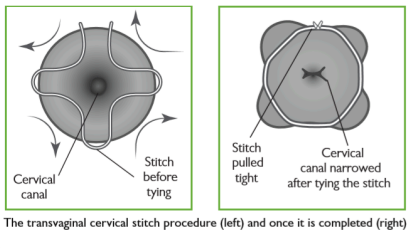
Hydatidiform mole (Ch 21 slide)

★ **PROFESSOR EMPHASIZED**
Why hCG is positive with no baby, why the uterus is too big, the follow-up hCG levels, and the cancer risk if follow-up is poor.

9.4 Premature cervical dilation (incompetent cervix)

2ND TRIMESTER | MCR: MODULE 9

PATHOPHYSIOLOGY	The cervix shortens and dilates **without contractions**, leading to loss.
WHO GETS A CERCLAGE	History of loss from painless dilation (not from labor).
MEDICAL MANAGEMENT	**Cerclage:** a stitch woven through the cervix and pulled tight to keep it closed. Abdominal cerclage clients need a cesarean.
REMOVAL	Transvaginal cerclage is cut at **37 weeks**. Remove early if labor starts or membranes rupture, or the cervix tears.



The transvaginal cervical stitch procedure (left) and once it is completed (right)

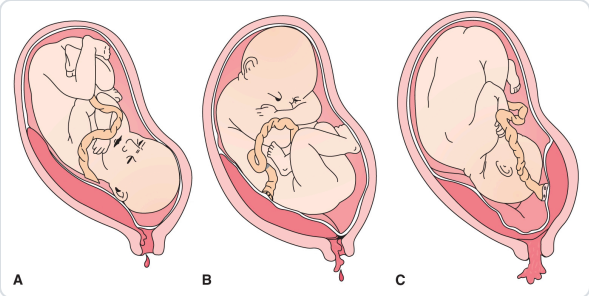
Transvaginal cerclage (Ch 21 slide)

CLIENT TEACHING	Take it easy; nothing in the vagina (no intercourse); no soaking in a tub; infection prevention (it is a suture).

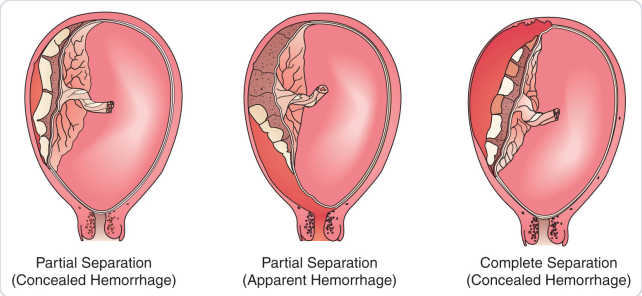
MY NOTES: CERCLAGE

9.5 Placenta previa vs placental abruption

3RD TRIMESTER · MCR: MODULE 9



Previa: (A) low-lying, (B) partial, (C) complete



Types of placental separation (abruption)

Previa is a **geography problem** (where the placenta implanted); abruption is the placenta **separating early** from the uterine wall. Tell them apart by bleeding color, pain, and how the uterus feels.

Feature	Placenta previa	Placental abruption
Bleeding	Painless, bright red	Dark red; can be concealed (nothing shows until delivery)
Pain	None or mild irritability	Sudden, severe abdominal pain
Uterus	Soft, relaxed	Hard, "board-like" (filling with blood), irritable contractions
Fetal heart	Usually reassuring, but check first	Normal, then drifts down, then bradycardia and late changes
Diagnosis	Ultrasound (usually the 20-week scan)	Clinical picture: an emergency
Risk factors	Grand multip, previous cesarean, advanced maternal age, uterine scarring (myomectomy, D&C, any uterine surgery), smoking, diabetes	Cocaine/amphetamines, hypertension, abdominal trauma, smoking, twins, a previous abruption
Do first	Assess bleeding and fetal heart tones; CBC, type and screen	Left side, oxygen, second large-bore IV, CBC, type and screen, coags
Do NOT	**No vaginal or cervical exams**, no intercourse, nothing in the vagina	Do not delay: the placenta is the baby's lifeline
Delivery	Complete previa = cesarean. A low-lying placenta at 20 weeks may move up; a complete one won't.	Emergent delivery (case: in the OR within 4 minutes); emptied uterus firms up and bleeding stops
Complications	Hemorrhage, fetal malposition, newborn with low hematocrit	Leading cause of maternal death from hemorrhage; DIC; can recur

★ **PROFESSOR EMPHASIZED**

Painless bright red = previa. Painful dark red + board-like abdomen = abruption. A subtle abruption showed up as the FHR baseline drifting down over hours (150 → 140 → 130) with bloody show. No vaginal exams with previa.

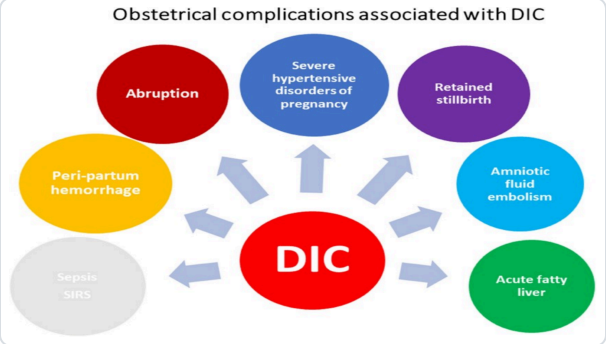
✎ **MEMORY TIP (FROM LECTURE)**

Previa is a **geography problem** (where it implanted). In abruption the uterus fills with blood and feels firm, "just like a hematoma would be firm if you touched it."

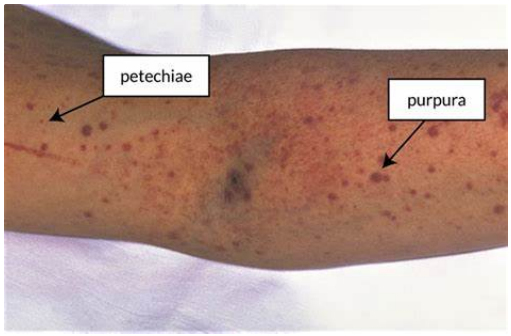
MY NOTES: PREVIA VS ABRUPTION

9.6 Disseminated intravascular coagulation (DIC)

EMERGENCY | MCR: MODULE 9



Obstetric complications associated with DIC (Ch 21 slide)



Petechiae and purpura

PATHOPHYSIOLOGY	Rare but life-threatening. After or during a hemorrhage, clotting activates everywhere and **uses up clotting factors and platelets**, so the blood can no longer clot and bleeding accelerates. Tiny clots also block flow to the liver, kidneys and brain.
OBSTETRIC CAUSES	Abruption; peripartum hemorrhage; severe hypertensive disorders (preeclampsia with severe features, HELLP); retained stillbirth / missed abortion with sepsis; amniotic fluid embolism; acute fatty liver.
CLINICAL MANIFESTATIONS	Bruising, petechiae, purpura, oozing from needle sites, nosebleeds, bleeding gums, bleeding anywhere, even after a hemorrhage seemed controlled.
LABS	Platelets **LOW**; fibrinogen **LOW**; PT/INR **HIGH**; PTT **HIGH**; D-dimer high; schistocytes (helmet cells) on smear.
NURSING PRIORITIES	Skin checks after any hemorrhage; volume expanders; blood replacement; keep her warm; ICU-level care.

★ **PROFESSOR EMPHASIZED**

Focus on **platelets and fibrinogen** (D-dimer is up in pregnancy anyway). Before any procedure, confirm the type and screen, CBC and coags were actually drawn (a retained-stillbirth case needed a mass transfusion with no type and screen on file).

MY NOTES: DIC

9.7 Preterm labor (PTL): terbutaline and betamethasone

MCR: MODULE 9

DEFINITION	Labor between 20 weeks and 36 weeks 6 days (slide: 20–37 weeks). Goal: stall or stop it if possible.
CAUSES & RISK FACTORS	**Infection** (UTI, STIs, BV, yeast: "a biggie"), dehydration, history of PTL, multiples, smoking, no prenatal care, PPROM, partner violence, accident or injury.
CUES (CASE: 31 WEEKS)	Regular contractions, pelvic pressure, nitrites in urine, cervix 3 cm / 50% at 31 weeks (should be closed).



Preterm infant (Ch 21 slide)

DIAGNOSTICS	Urinalysis and culture, CBC, type and screen, cervical cultures, fetal fibronectin (fFN): negative is reassuring; positive does **not** mean she will deliver.
NURSING CARE	Hydrate, treat the infection, activity restriction (evidence on strict bed rest and Trendelenburg is weak).

Medication	Purpose	Nursing points
Terbutaline (tocolytic)	Slows or halts contractions (off-label)	Boxed warning for maternal cardiac problems: short term only. Expect palpitations, tachycardia. Subcut every 30 min until contractions stop or HR 120. **Hold if HR over 120.**
Betamethasone (steroid, IM to mother)	Speeds fetal lung maturity (surfactant)	Usually 2 doses 24 hours apart; best 24–48 hours before birth (some benefit after 6 hours). Given **before 34 weeks** for any expected delivery.

★ PROFESSOR EMPHASIZED

Terbutaline is a **"band-aid" that buys time so betamethasone can work**. Start betamethasone as soon as possible. Know the 34-week threshold.

MY NOTES: PRETERM LABOR

9.8 Hypertensive disorders, preeclampsia, HELLP and magnesium

MCR: MODULE 9

Types: chronic hypertension, gestational hypertension, preeclampsia without severe features, preeclampsia with severe features, eclampsia, HELLP.

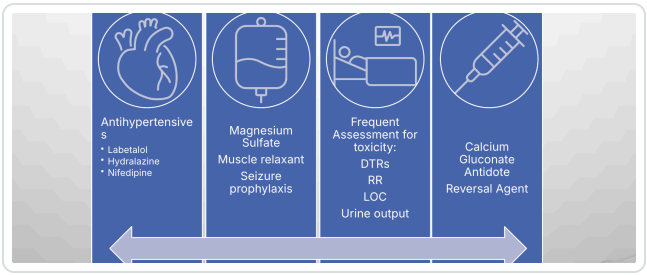
PATHOPHYSIOLOGY	Centered on the **placenta**: begins with pregnancy and resolves with delivery (not always right away; can be readmitted weeks postpartum). Narrow spiral arteries → vasospasm → hypoperfusion and endothelial injury. Usually 3rd trimester.

WHERE VASOSPASM HITS	Liver: epigastric pain, N/V. Kidney: low urine output, proteinuria, swelling. Brain: headache not relieved by Tylenol, hyperreflexia. Retina: blurred vision, spots. Lungs: pulmonary edema. Placenta: poor perfusion.
ASSESSMENT	BP, weight gain (case: 5 lb in 2 weeks), swelling of hands and face, epigastric pain, headache, visual changes, brisk DTRs, clonus, urine output, protein.
ECLAMPSIA	Preeclampsia plus seizures. Goal: prevent seizures.
SEVERE-RANGE BP	Systolic over 160 or diastolic over 110: every pregnancy-related stroke in a review had this. Give the antihypertensive within ~30 minutes (case BP 162/99).

HELLP	Complication of preeclampsia centered on the liver: **H**emolysis, **E**levated **L**iver enzymes (AST, ALT), **L**ow **P**latelets (under 100,000). Diagnosed by labs.
HELLP MANAGEMENT	Labs about every 8 hours. Rising AST/ALT and falling platelets = deliver. Platelets under 100,000 = no epidural or spinal.
ANTIHYPERTENSIVES	Labetalol, hydralazine, nifedipine; IV rescue for severe-range BP.



HELLP acronym (Ch 21 slide)



Preeclampsia treatments (Ch 21 slide): antihypertensives, magnesium, frequent toxicity checks, calcium gluconate

MAGNESIUM SULFATE	Seizure prophylaxis; muscle relaxant; also lowers BP. Expected: warm, flushed. Low stimulation (dim lights, few visitors, phone off), strict I&O, IV fluids no more than 125 mL/hr, VS every 30 min, continuous fetal monitoring.
MAG TOXICITY	Decreased or absent DTRs, RR down (e.g., 8), drowsy/decreased LOC, urine under 30 mL/hr, hypotension, headache, vision changes, slurred speech; can progress to cardiac arrest.
TOXICITY ACTIONS (IN ORDER)	**1. Stop the infusion. 2. Get a magnesium level. 3. Give calcium gluconate (antidote). 4. Notify the provider.** Can't get into the med dispenser? Bring the crash cart.
AFTER BIRTH	Magnesium relaxes the uterus: higher risk of postpartum hemorrhage and DIC.

★ PROFESSOR EMPHASIZED
You don't need fine detail on severe vs non-severe; understand preeclampsia as a whole (more or new symptoms, like visual changes, mean she is getting worse). The nurse does the frequent mag assessments: DTRs, RR, LOC, urine output.

✎ **MEMORY TIP (FROM LECTURE)**

"An allergic reaction to your own placenta." Mag toxicity: "**turn off the power before you start compressions**": stop the source first.

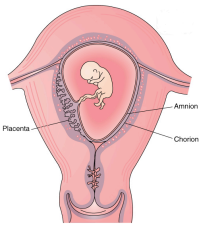
MY NOTES: PREECLAMPSIA / MAG

9.9 Amniotic fluid volume

MCR: MODULE 9

Condition	Volume	Think
Oligohydramnios	Less than 300 mL	Fetal **renal** problem
Polyhydramnios	More than 2000 mL	Fetal **GI** problem (not swallowing fluid)

Delivery risks with either: cord prolapse, postpartum hemorrhage, prolonged labor.



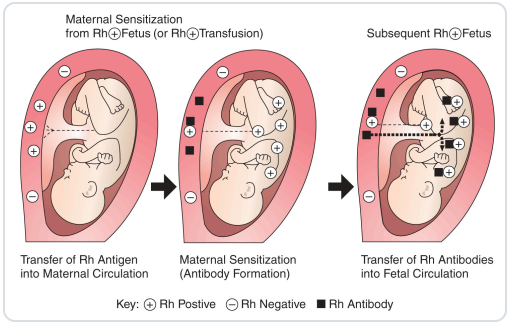
Amniotic fluid (Ch 21 slide)

★ **PROFESSOR EMPHASIZED**

Review concept: know the cutoffs and which fetal system each points to.

9.10 Rh isoimmunization and RhoGAM

MCR: MODULE 9



Rh sensitization: first exposure, antibody formation, later pregnancy affected (Ch 21 slide)

PATHOPHYSIOLOGY	An Rh-negative mother exposed to Rh-positive fetal blood makes antibodies that attack Rh-positive fetuses in **later** pregnancies. The baby's type is checked from cord blood after birth; the mother always gets a type and screen.
RHOGAM	Synthetic antibody that keeps her from making her own (it "tricks" her body); does not harm the baby.
WHEN	**28 weeks**; **within 72 hours of birth** if the baby is Rh positive; any time blood may mix: miscarriage, D&C, abortion, ectopic, amniocentesis, CVS, abdominal trauma.
TESTING	Coombs test checks whether antibodies have formed.

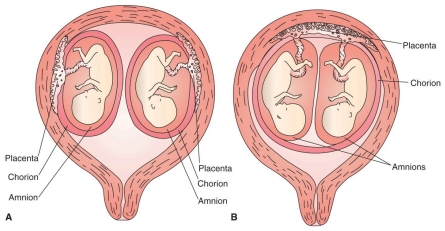
★ **PROFESSOR EMPHASIZED**

Why, when, and what it protects: **subsequent pregnancies**.

9.11 Multiple gestation

MCR: MODULE 9

Type	How	Placenta
Dizygotic (fraternal)	Two eggs + two sperm; same or different sex	Each has its own
Monozygotic (identical)	One egg + one sperm that splits	Shared: higher risk (twin-to-twin transfusion)



Twin placentation (Ch 21 slide)

★ PROFESSOR EMPHASIZED

Only fraternal vs identical detail is needed. Twins carry more risk overall: preterm birth and one twin smaller than the other.

MY NOTES: RH / FLUID / TWINS

9.12 Module 9 practice: check yourself

NGN-STYLE · ANSWERS ON THE NEXT PAGE

CASE
A G2P1 calls the unit at 9 weeks: "I think I'm miscarrying." Later items use other lecture cases (ectopic with shock signs, 31-week preterm labor, preeclampsia on magnesium).

M9-1 · MC · take action **What is the nurse's best response to the caller?**

A. "Lie down on bed rest for 24 hours."
B. "Spotting is harmless; keep up light activity."
C. "Come in to be examined, and save any tissue you pass."
D. "Use a tampon to put pressure on the cervix."

M9-2 · Matrix · analyze cues **Match each presentation to the type of spontaneous abortion.**

Finding	Threatened	Imminent	Complete	Missed
Cervix closed, FHT present, brown spotting, no pain	○	○	○	○
Cervix dilated, membranes bulging, contracting	○	○	○	○
All tissue passed, no FHT	○	○	○	○
No FHT on scan, no labor signs, cervix unchanged	○	○	○	○

M9-3 · SATA · recognize cues **Which history items raise the risk of ectopic pregnancy? Select all that apply.**

A. Pelvic inflammatory disease
B. IUD in place
C. Previous tubal ligation
D. Two previous vaginal births
E. Endometriosis

M9-4 · MC · prioritize **A client at 7 weeks has sudden left lower quadrant pain. BP 95/58, HR 132, RR 24, pale and diaphoretic. What does the nurse anticipate?**

A. Methotrexate and discharge
B. Emergency surgery for a ruptured ectopic
C. An ultrasound next week
D. Bed rest and serial hCG

M9-5 · SATA · teaching **Which instructions belong in methotrexate teaching? Select all that apply.**

A. Stop prenatal vitamins and folic acid
B. Avoid alcohol
C. Expect serial hCG levels
D. Try to conceive again right away
E. Wait a few months before conceiving

M9-6 · MC · recognize cues **At 13 weeks by dates, the fundus is near the umbilicus, hCG is very high, and she has dark brown bleeding. Which condition fits?**

A. Placenta previa
B. Molar pregnancy
C. Threatened abortion
D. Multiple gestation with twin-to-twin transfusion

M9-7 · Matrix · analyze cues Mark each finding as previa or abruption.

Finding	Previa	Abruption
Painless bright red bleeding	○	○
Board-like, tender uterus	○	○
Dark red bleeding after cocaine use	○	○
Soft, relaxed uterus	○	○
Seen on the 20-week ultrasound	○	○

M9-8 · MC · take action **A client with a known complete previa is admitted for bleeding. Which order should the nurse question?**

A. CBC and type and screen
B. Continuous fetal monitoring
C. Sterile vaginal exam to check dilation
D. Saline lock

M9-9 · SATA · analyze cues **After an abruption, which labs suggest DIC? Select all that apply.**

A. Platelets low
B. Fibrinogen low
C. PT/INR prolonged
D. Fibrinogen high
E. Schistocytes on the smear

M9-10 · MC · take action **A 31-week client in preterm labor is due for terbutaline. Her HR is 126. What does the nurse do?**

A. Give the dose
B. Hold the dose and report
C. Give half the dose
D. Give it IM instead

M9-11 · MC · evaluate **Which statement shows she understands betamethasone?**

A. "It will stop my contractions."
B. "It helps my baby's lungs mature before birth."
C. "It prevents seizures."
D. "It treats my infection."

M9-12 · SATA · recognize cues **A client on magnesium sulfate: which findings suggest toxicity? Select all that apply.**

A. Absent DTRs
B. RR 8
C. Urine output 20 mL/hr
D. Feeling warm and flushed
E. Drowsy, slurred speech

M9-13 · Ordering · take action **Put the magnesium toxicity actions in order:**

A. Give calcium gluconate
B. Stop the magnesium infusion
C. Notify the provider
D. Obtain a magnesium level

M9-14 · MC · prioritize **A preeclamptic client's BP is 162/99. What is the priority?**

A. Recheck in 4 hours
B. Give the ordered antihypertensive within about 30 minutes
C. Encourage ambulation
D. Offer acetaminophen

M9-15 · MC · analyze cues **Which lab trend in HELLP means delivery is indicated?**

A. Rising platelets, falling AST
B. Rising AST/ALT and falling platelets
C. Falling hemoglobin only
D. Normal liver enzymes

M9-16 · MC · take action **An Rh-negative client has a D&C after a miscarriage at 10 weeks. What does she need?**

A. Nothing; it's too early
B. RhoGAM
C. A Coombs test on the fetus
D. Betamethasone

M9-17 · MC · analyze cues **An ultrasound shows polyhydramnios. Which fetal system does the nurse think about?**

A. Renal
B. GI (swallowing)
C. Cardiac
D. Neurologic

9.13 Module 9 answer key & rationales

CHECK AFTER YOU TRY

M9-1 **C**
Only an exam gives information. Bed rest doesn't stop a loss, spotting isn't harmless to dismiss, and a tampon hides blood you need to quantify.

M9-2 **Threatened, Imminent, Complete, Missed (in order)**
Sort by how much tissue has been lost: closed cervix with FHT = threatened; dilated with bulging membranes = imminent; all passed = complete; no FHT and no signs = missed.

M9-3 **A, B, C, E**
Anything that scars or blocks the tube (PID, ligation, endometriosis) or prevents uterine implantation (IUD).

M9-4 **B**
One-sided pain plus shock signs = ruptured ectopic, a surgical emergency. Methotrexate is for unruptured.

M9-5 **A, B, C, E**
Folic acid supports cell growth (we want cell death); alcohol can cause severe vomiting; serial hCG confirms it drops; wait so hCG is reliable.

M9-6 **B**
Uterus larger than dates, very high hCG, and dark brown "sludgy" bleeding are classic molar pregnancy.

M9-7 **Previa: 1, 4, 5 · Abruption: 2, 3**
Painless bright red, soft uterus, seen on ultrasound = previa. Board-like uterus, dark red, stimulants = abruption.

M9-8 **C**
No vaginal or cervical exams with previa: you can hit the placenta and cause hemorrhage.

M9-9 **A, B, C, E**
DIC uses up platelets and fibrinogen, so both are low and clotting times are prolonged. Fibrinogen is not high.

M9-10 **B**
Hold terbutaline if the heart rate is over 120 (boxed warning for maternal cardiac problems).

M9-11 **B**
Betamethasone speeds fetal lung maturity (surfactant). Terbutaline slows contractions; magnesium prevents seizures.

M9-12 **A, B, C, E**
Warm and flushed is an expected effect, not toxicity.

M9-13 **Stop → level → calcium gluconate → provider**
"Turn off the power before you start compressions": stop the source first.

M9-14 **B**
Systolic over 160 or diastolic over 110 is severe range (stroke risk); treat within about 30 minutes.

M9-15 **B**
Rising liver enzymes and falling platelets mean HELLP is worsening: deliver. Under 100,000 platelets = no epidural.

M9-16 **B**
RhoGAM any time maternal and fetal blood may mix, including miscarriage and D&C, to protect future pregnancies.

M9-17 **B**
Polyhydramnios (over 2000 mL) points to a GI problem (not swallowing); oligohydramnios points to the kidneys.

MODULE 10 · CHAPTERS 23 & 24

Complications of Labor & Birth · Surgical Interventions

Organized by the P's: Power (contractions), Passenger (fetal position, presentation, size, cord), Passage (the birth canal). Dystocia = any deviation that impedes labor progress (dilation or descent), fast or slow.

10.1 Group B Streptococcus (GBS)

INFECTION · MCR: MODULE 10

WHAT IT IS	Normal flora of the vaginal and rectal area. **Not** an STI and not contagious to others, but the baby is exposed in vaginal birth and can develop sepsis, meningitis or pneumonia.
TESTING	Routine swab at about **36 weeks** (sooner in preterm labor). Done on everyone eligible for vaginal birth, even with a planned cesarean. Retested every pregnancy. Also treat if GBS grows in urine.
TREATMENT	Antibiotics in labor: **penicillin**, doses 4 hours apart. Goal: **at least 2 doses before birth** = covered. Provider decides when to start.
IF NOT COVERED	Increased newborn surveillance for infection; treat the baby if signs appear.

MY NOTES: GBS

10.2 Power problems and dysfunctional labor

POWER · MCR: MODULE 10

Feature	Hypotonic contractions	Hypertonic contractions
What	Weak, infrequent, uncoordinated ("wimpy")	Too frequent, too much tone, no rest between; painful but ineffective
Effect	Longer labor	Exhausting for mom; uterus and placenta can't refill, so less oxygen to the baby, then bradycardia
Causes	Full bladder; over-distended uterus (multiples, polyhydramnios, macrosomia, grand multip)	Latent phase with an unripe cervix while oxytocin is running (too much oxytocin)
Treatment	Oxytocin (augmentation), amniotomy	**Turn off oxytocin FIRST**; fluids; position change; terbutaline only if it persists

Stage	Problem	Causes
First stage = dilation	Prolonged latent or active phase; arrest of dilation (e.g., stuck at 7 cm for 4–5 hours despite oxytocin, amniotomy, walking)	Unripe cervix, stuck in bed, CPD, large baby
Second stage = descent	Prolonged descent; arrest of descent (e.g., stuck at +2 for 3 hours); arrest usually means cesarean	OP position, CPD, pelvic shape

DYSTOCIA INTERVENTIONS (SLIDE)	Hydration, pain relief, less stimulation (lights, noise), cervical ripening, oxytocin induction or augmentation, amniotomy, cesarean if needed.

★ **PROFESSOR EMPHASIZED**

Keep bladders empty in labor and postpartum: a full bladder blocks descent, stops the uterus contracting, and raises hemorrhage risk. Goal pattern: a contraction every 2–3 minutes with at least 60 seconds of rest. Keep stages simple: first stage = dilation, second stage = descent. You don't need the hour cutoffs, just what "arrest" means.

MY NOTES: POWER / DYSTOCIA

10.3 Precipitous birth

POWER

DEFINITION	Birth within **3 hours** of contractions starting.
MATERNAL RISKS	Lacerations (perineum doesn't stretch gradually), hemorrhage, placental abruption.
NEWBORN RISKS	Head trauma (no time to mold), bleeding (fragile vessels, no vitamin K yet), respiratory distress (not squeezed, more fluid), looks "stunned."
NURSING & TEACHING	Often nurse-assisted; straight into a labor bed. Teach that the next birth may be even faster: come in as soon as contractions begin.

10.4 Induction, augmentation, Bishop score and cervical ripening

MCR: MODULE 10

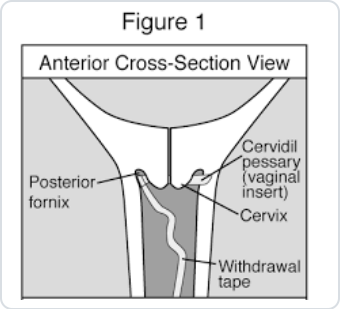
	Induction	Augmentation
Definition	Artificially starting labor	Speeding up labor that started on its own but stalled (hypotonic)
Reasons	Preeclampsia, GDM, post-dates (40–42+ weeks), IUGR, other medical conditions	Weak, spaced, irregular contractions; ROM with stalled contractions (infection risk)
Tools	Ripening if Bishop is low → oxytocin, amniotomy	Oxytocin, amniotomy

Modified Bishop score (Ch 23 slide): how inducible is she? Five parts, 0–3 points each. **8 or more = ripe** enough to go ahead with oxytocin.

Component	0	1	2	3
Dilation (cm)	Closed	1–2	3–4	5 or more
Effacement (%)	0–30	40–50	60–70	80 or more
Station	−3	−2	−1, 0	+1, +2
Consistency	Firm	Medium	Soft	
Position	Posterior	Mid	Anterior	

Lecture example: 2 cm (1) + 25% (0) + −2 (1) + medium (1) + posterior (0) = **3**, so she needs a ripening agent.

DINOPROSTONE (CERVIDIL / PREPIDIL)	Prostaglandin wafer behind the cervix (posterior fornix), releases over 12 hours. **Withdrawal tape**: pull it for hypertonic contractions, fetal distress or tachysystole. Teach: don't pull the string; stay in bed a few hours.
MISOPROSTOL (CYTOTEC)	Prostaglandin tablets, buccal or vaginal.
FOLEY BULB / COOK CATHETER	Mechanical balloon dilates the cervix and stimulates natural prostaglandins; non-drug; painful but effective.



Cervidil insert with withdrawal tape (Ch 23 slide)

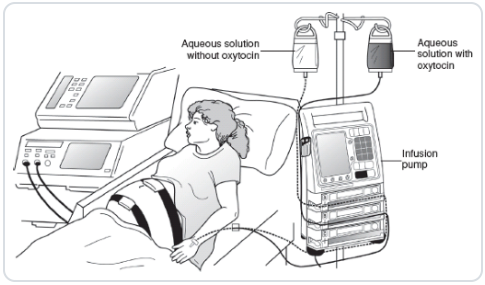
★ PROFESSOR EMPHASIZED
You don't memorize the Bishop chart; you **apply it to a case**. Every ripening method's goal is to stimulate prostaglandins. Order: ripening agent, then oxytocin. Remove prostaglandins for fetal distress or tachysystole.

✎ MEMORY TIP (FROM LECTURE)
Cervix consistency: firm = tip of your nose, medium = ear lobe, soft = inside of your cheek.

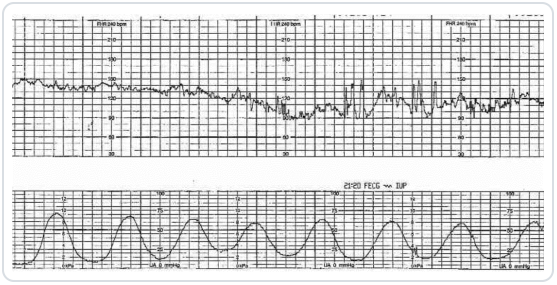
MY NOTES: INDUCTION / BISHOP

10.5 Oxytocin administration

HIGH-ALERT DRUG · MCR: MODULE 10



Infusion setup (Ch 23 slide)


Contraction tracing (Ch 23 slide)

VERIFY BEFORE STARTING	Head engaged; vertex, longitudinal lie; cervix ripe (documented exam, or advocate for ripening); no CPD; reassuring strip (category I or II); no contraindication to vaginal birth (e.g., active lesions).
SETUP	Its own pump/channel, **not piggybacked**; lactated Ringer's main line; connect at the port **closest to the client** so a fluid bolus doesn't push oxytocin.
MONITORING	Continuous fetal and contraction monitoring; document every 30 minutes; watch for water intoxication.
TITRATION	Contractions 5 min apart: increase. 2–3 min apart: leave it. Unstable fetal heart rate: reduce or **turn it off**.

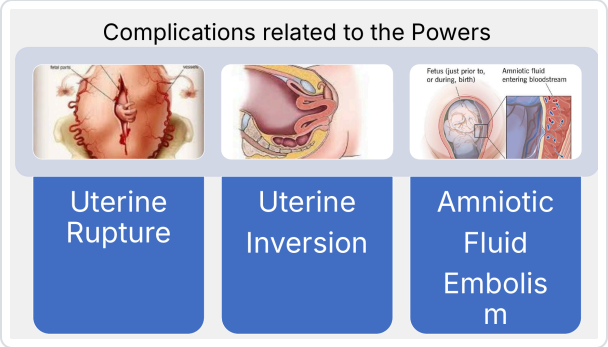
STOP FOR	More than 5 contractions in 10 minutes, a contraction over 2 minutes, or a non-reassuring fetal heart rate. Risks: uterine rupture, fetal hypoxia.

★ PROFESSOR EMPHASIZED
Oxytocin is on the national high-alert list and is in nearly every OB legal case. **"It is not a race."** Be conservative; a sloppy pattern left running hurts the baby.

MY NOTES: OXYTOCIN

10.6 Uterine rupture, uterine inversion, amniotic fluid embolism

EMERGENCIES · MCR: MODULE 10



Complications related to the powers (Ch 23 slide)

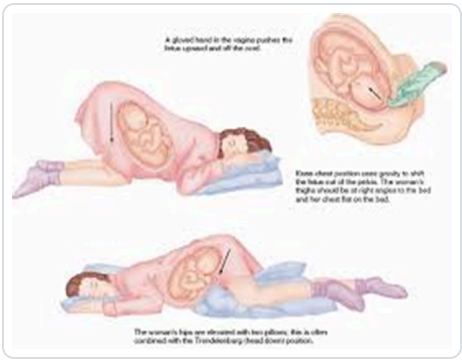
Condition	What	Cues / risks	Actions
Uterine rupture	The uterus opens; fetus may be in the abdomen	**Sudden excruciating pain** out of proportion. Risks: previous cesarean scar, hypertonic contractions or too much oxytocin, thinned muscle, trauma	OR immediately, oxygen, fluids, blood
Uterine inversion	Uterus turns inside out (NOT normal)	Excessive bleeding. Causes: too much fundal pressure, pulling on the cord in third stage	Provider replaces the uterus; treat hemorrhage
Amniotic fluid embolism	Fluid enters maternal circulation, blocking/constricting the pulmonary artery	Sudden dyspnea, cyanosis, unconsciousness, cardiac arrest, then DIC. Rare, unpredictable, high mortality; CPR less effective	Emergency response; DIC care

★ PROFESSOR EMPHASIZED
These are rare, but nurses must recognize them. Inversion is the abnormal one.

10.7 Umbilical cord prolapse

PASSENGER · EMERGENCY · MCR: MODULE 10

PATHOPHYSIOLOGY	The cord slips below the presenting part and the head compresses it: little or no blood flow to the baby.
RISK FACTORS	AROM with a high station (head isn't sealing the cervix), prematurity or small baby, polyhydramnios, transverse lie, long cord. Ruptured with a high station: don't get her up.
RECOGNIZE	Sudden fetal bradycardia after membranes rupture; cord felt on exam (case: 33 weeks, AROM, FHR drops to 90).



Knee-chest and Trendelenburg-type positions (Ch 23 slide)

ACTIONS (IN ORDER)	**1. Lift the presenting part off the cord with a gloved hand and keep it there; do not let go** until the OR. 2. Call for help ("we have a cord"). 3. Oxygen. 4. Prepare OR and anesthesia for emergency cesarean. 5. If delayed: knee-chest or hands-and-knees (temporary only); cover an exposed cord with a saturated sterile towel.
AFTER	Heart rate usually recovers once the head is lifted, but she still needs a cesarean.

★ PROFESSOR EMPHASIZED

Lift the head off the cord and **never let go**. Positioning is a band-aid. Any ruptured client with a high station is a prolapse risk.

✎ MEMORY TIP (FROM LECTURE)

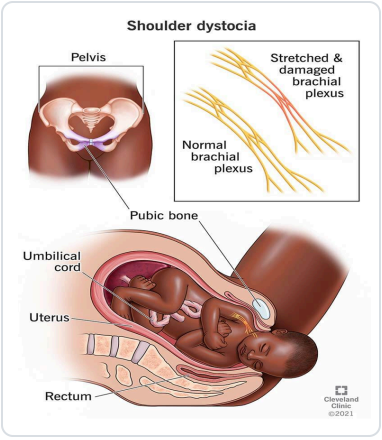
A kinked garden hose: keep the pressure on, no flow gets through.

MY NOTES: CORD PROLAPSE

10.8 Shoulder dystocia and macrosomia

PASSENGER · MCR: MODULE 10

WHAT IT IS	Head delivers, but the anterior shoulder is stuck under the pubic bone. About **5 minutes** before serious hypoxia.
RISK FACTORS	GDM, **macrosomia (over 4000 g)**, post-dates; can still happen with a small baby or small pelvis.
NURSE'S ROLE	Document the time the head delivers and the time the body delivers. Call for help and the NICU team.
INTERVENTIONS	**McRoberts** (legs pulled all the way back, knees to chest). **Suprapubic pressure** when directed. **NO fundal pressure.** Provider may rotate the baby, deliver the other arm, or break the clavicle.



Shoulder dystocia (Ch 23 slide)

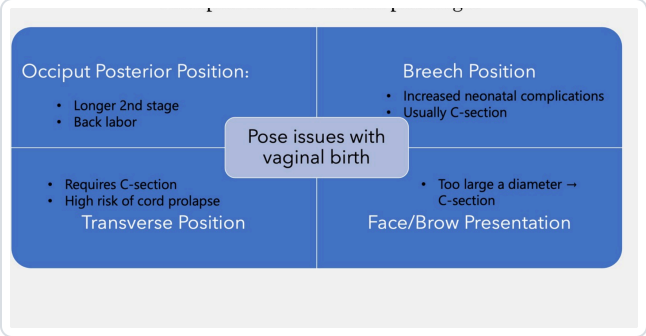
NEWBORN	Brachial plexus injury (Erb's palsy): check arm movement and tone. Clavicle fracture: palpate for crepitus. Be ready to resuscitate.

★ PROFESSOR EMPHASIZED
McRoberts first, suprapubic (never fundal) pressure, time head-to-body, assess the arm and clavicle after birth.

MY NOTES: SHOULDER DYSTOCIA

10.9 Malpresentation (posterior, breech) and external cephalic version

PASSENGER · MCR: MODULE 10



Complications with the passenger (Ch 23 slide)

Position	Concern	Management
Occiput posterior (OP)	Longer second stage, back labor, more rotation	Vaginal birth possible, just harder
Breech	More neonatal complications and birth injury	Usually cesarean
Transverse	Can't deliver vaginally; high cord-prolapse risk	Cesarean
Face / brow	Too large a diameter	Cesarean

ECV	Scheduled procedure (not in labor) to turn a breech or transverse baby from outside the abdomen.
ECV REQUIREMENTS	37+ weeks; adequate fluid (not oligohydramnios); single fetus; reactive NST.
ECV NURSING	Empty bladder; ultrasound in the room to confirm position; monitor fetal heart tones; pain meds or spinal; terbutaline to relax the uterus; keep her off her back (towel under the right hip).

10.10 The passage: CPD and inlet/outlet contraction

PASSAGE · MCR: MODULE 10

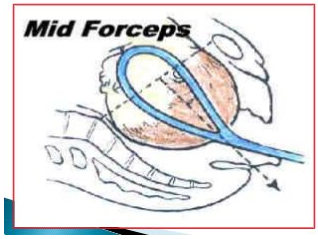
CPD	Cephalopelvic disproportion: the pelvis is too small for the baby's head.
INLET / OUTLET CONTRACTION	Inlet: AP diameter under 11 cm or transverse under 12 cm. Causes include genetics and childhood bone disease such as rickets. Found by pelvimetry (provider's manual exam).

TRIAL OF LABOR	Borderline measurements with good position: she may try. Arrest of dilation or descent = cesarean. Later pregnancies may be scheduled cesareans.

10.11 Assisted vaginal birth: forceps and vacuum

MCR: MODULE 10

	Forceps	Vacuum
Device	Spoon-like blades on each side of the head, joined in the middle	Suction cup with a gauge (green/yellow/red); pops off if pressure is too high
Baby risks	Bruising, lacerations, facial nerve involvement	Cephalohematoma (resolves on its own), scalp injury
Attempts	Up to 3	Up to 3 pop-offs, then stop



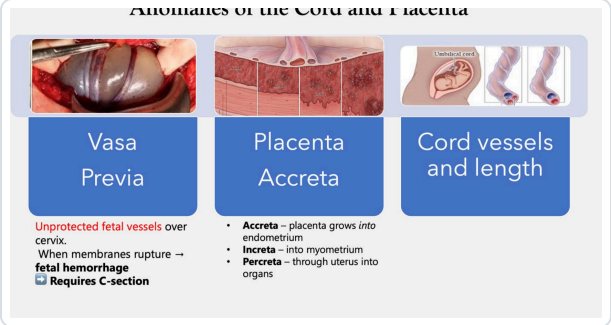
Mid forceps (Ch 23 slide)

BEFORE HANDING IT TO THE PROVIDER	Empty bladder; **fully dilated** and effaced (a 9 cm forceps case ended in a major hemorrhage); membranes ruptured; head engaged.

✎ **MEMORY TIP (FROM LECTURE)**
Forceps are like salad tongs.

10.12 Cord and placenta anomalies

MCR: MODULE 10



Anomalies of the cord and placenta (Ch 23 slide)

Condition	What it is	Key points
Vasa previa	Unprotected fetal vessels run through the membranes over the cervix	When membranes rupture the vessels tear: **fetal** hemorrhage, sudden bradycardia, red blood with the fluid. Ultrasound diagnosis; surveillance; scheduled cesarean, never labor
Accreta / increta / percreta	Accreta = into the endometrium; increta = into the myometrium; percreta = through the uterus into other organs	Seen on ultrasound; major hemorrhage risk; percreta may need cesarean hysterectomy
Velamentous insertion	Cord vessels branch in the membranes before reaching the placenta	Vulnerable if membranes tear
Two-vessel cord	Normal = 2 arteries + 1 vein; one artery missing	Found at delivery; work up for cardiac and renal anomalies

Succenturiate lobe / battledore	Placenta in two lobes / cord inserts at the margin

★ **PROFESSOR EMPHASIZED**
Vasa previa bleeding is **fetal blood**. Know accreta vs increta vs percreta. A two-vessel cord needs a work-up.

MY NOTES: CORD / PLACENTA

10.13 Amniotomy (AROM)

CH 24 · MCR: MODULE 10

WHAT & WHY	Artificial rupture of membranes with an amniohook; speeds labor and stimulates prostaglandins and oxytocin. Often part of induction; she may decline.
BEFORE	Head engaged and applied to the cervix; stable fetal heart rate; adequate dilation.
NURSE	**Fetal heart tones before and after** (most important). Document color, odor, amount and time. Meconium: be ready at delivery.
RISKS	Umbilical cord prolapse; amniotic fluid embolism.

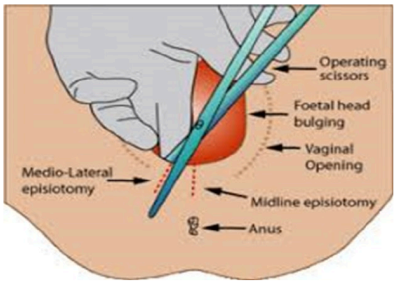


Amniohook (Ch 24 slide)

10.14 Episiotomy

CH 24 · MCR: MODULE 10

WHAT	Incision of the perineum in second stage to enlarge the opening. **Not routine**; only as needed. The consent lists possible episiotomy and lacerations because there may be no time to discuss it in an emergency.
INDICATIONS	Macrosomic infant; preterm infant (less pressure on the head); shortening second stage in emergencies such as fetal distress.



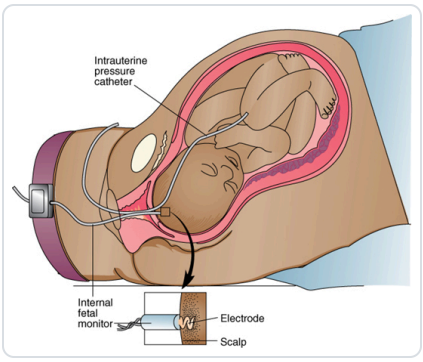
Midline vs mediolateral (Ch 24 slide)

Type	Direction	Advantages	Disadvantages
Midline	Straight down toward the rectum	Heals well, less pain	Higher risk of 3rd–4th degree tears into the rectum; can result in a retroperitoneal hematoma (slide)
Mediolateral	Downward at an angle	Lower risk of rectal extension	More discomfort, harder repair

10.15 Internal monitoring (FSE, IUPC) and scalp stimulation

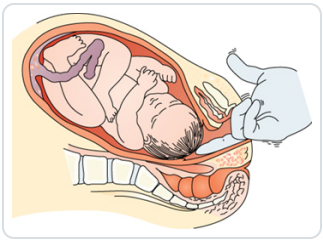
CH 24 · MCR: MODULE 10

FETAL SCALP ELECTRODE (FSE)	Spiral electrode on the fetal scalp: continuous, accurate fetal heart rate.
IUPC	Catheter in the uterus: most accurate contractions, exact strength in mmHg and Montevideo units. Port allows amnioinfusion (fluid to cushion the cord and relieve variable decelerations).
REASONS	Obesity, high-risk pregnancy, trouble tracing externally, concerning strip. Not standard for everyone.
REQUIREMENTS	Membranes **ruptured** and at least **3 cm** dilated.
DISADVANTAGES	Infection; limited mobility.



IUPC and FSE (Ch 24 slide)

SCALP STIMULATION	Provider presses or scratches the fetal head during a vaginal exam. **Expected: an acceleration** (reassuring). No acceleration may mean acidosis or poor oxygenation: possible cord gas or change in plan.



Scalp stimulation (Ch 24 slide)

10.16 Cesarean birth

CH 24 · MCR: MODULE 10

Type	Indications
Scheduled	Known CPD, malpresentation, active HSV lesions, repeat cesarean, previous classical incision, known previa
Elective	Client request; discouraged (major surgery); counsel on risks; insurance may not pay
Emergency	Fetal distress, abruption, cord prolapse, failed vacuum or forceps, arrest of descent

Incision	Notes
Classical (vertical)	High-contractility area: higher rupture risk, even laboring at home. **Not eligible for vaginal birth after cesarean**; repeat cesareans.
Low transverse	Low-contractility area: lower rupture risk.

MATERNAL RISKS	More blood loss and postpartum hemorrhage; infection; long recovery; organ injury (bladder); scarring that complicates future pregnancies.
NEWBORN RISKS	Respiratory difficulty (not squeezed, more lung fluid); temperature (cold OR).
SCHEDULED NURSING	NPO after midnight; teaching (she'll be behind a curtain); IV; prep. Nurse circulates, anesthesia gives meds, two providers operate.

★ **PROFESSOR EMPHASIZED**

Classical incision = no vaginal birth later. Know maternal risks. Elective cesarean needs risk counseling.

10.17 Module 10 practice: check yourself

NGN-STYLE · ANSWERS ON THE NEXT PAGE

CASE
A G1 at 40 weeks 5 days is admitted for induction. GBS positive at 36 weeks. (Later items use the lecture cases: cord prolapse at 33 weeks, shoulder dystocia, internal monitoring.)

M10-1 · MC · evaluate **When is she considered adequately covered for GBS?**

A. One dose of penicillin any time in labor
B. At least 2 doses of penicillin, 4 hours apart, before birth
C. Antibiotics started after birth
D. A single dose at 36 weeks

M10-2 · Calculation · analyze cues **Her exam: 3 cm, 50% effaced, station −1, soft, mid-position. Using the Bishop chart, what is her score, and is she ready for oxytocin?**

M10-3 · SATA · recognize cues **Before starting oxytocin, the nurse verifies which? Select all that apply.**

A. Fetal head engaged, vertex
B. Ripe cervix documented
C. Category I or II strip
D. Known CPD
E. Oxytocin piggybacked into the port closest to the client

M10-4 · MC · take action **On oxytocin, she has 6 contractions in 10 minutes and the FHR is falling. First action?**

A. Increase the rate
B. Turn off the oxytocin
C. Call for an epidural
D. Do a vaginal exam

M10-5 · MC · take action **A dinoprostone insert is in place and tachysystole develops. What does the nurse do?**

A. Leave it; it lasts 12 hours
B. Remove it with the withdrawal tape
C. Start oxytocin
D. Give misoprostol

M10-6 · MC · prioritize **A 33-week client has AROM; the FHR drops to 90 and the nurse feels the cord. First action?**

A. Place her in knee-chest
B. Lift the presenting part off the cord and keep it there
C. Apply oxygen
D. Call the provider from the desk

M10-7 · MC · take action **During a shoulder dystocia, which action should the nurse NOT do?**

A. McRoberts maneuver
B. Suprapubic pressure when directed
C. Fundal pressure
D. Note the time the head delivered

M10-8 · SATA · recognize cues **Which risk factors predispose to shoulder dystocia? Select all that apply.**

A. Gestational diabetes
B. Macrosomia over 4000 g
C. Post-dates
D. Oligohydramnios
E. Preterm birth

M10-9 · MC · analyze cues **A laboring client with a previous cesarean suddenly has excruciating pain out of proportion to labor. What does the nurse suspect?**

A. Placenta previa
B. Uterine rupture
C. Normal transition
D. Vasa previa

M10-10 · SATA · generate solutions **Requirements before an external cephalic version? Select all that apply.**

A. 37 weeks or more
B. Adequate amniotic fluid
C. Twin gestation
D. Reactive NST
E. Active labor

M10-11 · Matrix · take action **Before a vacuum-assisted birth, which conditions must be met?**

Finding	Required	Not required
Fully dilated	○	○
Membranes ruptured	○	○
Empty bladder	○	○
Head engaged	○	○
Epidural in place	○	○

M10-12 · MC · analyze cues **Membranes rupture and there is red blood in the fluid with sudden fetal bradycardia. Whose blood is it most likely?**

A. Mother's, from placenta previa
B. The fetus's, from vasa previa
C. Mother's, from a cervical laceration
D. Bloody show

M10-13 · MC · prioritize **What is the most important nursing action with an amniotomy?**

A. Document the time only
B. Assess fetal heart tones before and after
C. Ambulate her right away
D. Start oxytocin

M10-14 · SATA · recognize cues **Which are true about internal monitoring? Select all that apply.**

A. Membranes must be ruptured
B. Cervix at least 3 cm
C. IUPC shows exact contraction strength
D. No infection risk
E. Readings stay adequate when she moves

M10-15 · MC · evaluate **After scalp stimulation, which response is reassuring?**

A. FHR acceleration
B. No change
C. Late deceleration
D. Bradycardia

M10-16 · MC · teaching **A client with a previous classical cesarean asks about a vaginal birth next time. Best answer?**

A. "You can try once labor starts."
B. "A classical incision isn't eligible for vaginal birth; you'll have a repeat cesarean."
C. "Only if the baby is small."
D. "Yes, with an epidural."

M10-17 · MC · take action **Hypotonic contractions; her bladder is distended. First action?**

A. Start oxytocin
B. Help her empty her bladder
C. Perform amniotomy
D. Prepare for cesarean

10.18 Module 10 answer key & rationales

CHECK AFTER YOU TRY

M10-1 **B**
Penicillin every 4 hours; at least 2 doses before birth = covered. Otherwise the newborn gets increased surveillance.

M10-2 **2 + 1 + 2 + 2 + 1 = 8 → yes**
A score of 8 or more means the cervix is ripe enough to proceed with oxytocin. Apply the chart; don't memorize it.

M10-3 **A, B, C**
CPD is a contraindication; oxytocin runs on its own pump, connected at the port closest to the client (not piggybacked).

M10-4 **B**
More than 5 contractions in 10 minutes with a falling FHR: turn off oxytocin first, then fluids and position.

M10-5 **B**
The withdrawal tape lets you remove prostaglandins for tachysystole or fetal distress.

M10-6 **B**
Relieve cord compression first and don't let go; then call for help, oxygen, and OR. Knee-chest is a temporary band-aid.

M10-7 **C**
Never fundal pressure; it wedges the shoulder further. McRoberts and suprapubic pressure create space.

M10-8 **A, B, C**
GDM, macrosomia and post-dates. It can still happen with a small baby.

M10-9 **B**
Sudden, out-of-proportion pain with a uterine scar = rupture: OR immediately, oxygen, fluids, blood.

M10-10 **A, B, D**
Also a single fetus; ECV is a scheduled procedure, not done in labor.

M10-11 **Required: fully dilated, ruptured, empty bladder, head engaged**
An epidural is not on the list. Full dilation matters (a 9 cm forceps case ended in major hemorrhage).

M10-12 **B**
Vasa previa: unprotected fetal vessels tear when membranes rupture, so the bleeding is fetal.

M10-13 **B**
FHT before and after is the most important step (watch for cord prolapse); then document color, odor, amount and time.

M10-14 **A, B, C, E**
Disadvantages are infection risk and limited mobility.

M10-15 **A**
An acceleration is the expected, reassuring response; none may suggest acidosis.

M10-16 **B**
Classical (vertical) incisions sit in a high-contractility area with higher rupture risk.

M10-17 **B**
A full bladder blocks descent and weakens contractions; empty it before escalating.

R Rapid review: priority actions and medications

LAST LOOK

Situation	Do this
Magnesium toxicity	Stop infusion → mag level → calcium gluconate → provider
Oxytocin + tachysystole or bad strip	Turn off oxytocin first; fluids; position; terbutaline if it persists
Prostaglandin insert + tachysystole/distress	Remove it (withdrawal tape)
Cord prolapse	Lift presenting part off the cord, never let go; help; O2; OR. Knee-chest is temporary
Shoulder dystocia	McRoberts; suprapubic pressure; NO fundal pressure; time head-to-body
Suspected previa	Bleeding + FHT, CBC, type and screen; NO vaginal exam
Suspected abruption	Left side, O2, second large-bore IV, CBC/T&S/coags, FHT
Uterine rupture	OR immediately, O2, fluids, blood
Amniotic fluid embolism	Emergency response; DIC care
Uterine inversion	Provider replaces uterus; treat hemorrhage
Ruptured ectopic	Treat shock; surgery
Severe-range BP (≥160/≥110)	Antihypertensive within ~30 minutes
Orthopnea (cardiac)	Sit her up; never supine
Newborn of diabetic mother	Feed within 30 min; heel-stick glucose

Medication	Key points
Terbutaline	Tocolytic; short term; hold if HR > 120
Betamethasone	IM to mother; fetal lungs; 2 doses 24 h apart; before 34 weeks
Magnesium sulfate	Seizure prophylaxis; ≤125 mL/hr IV fluids; VS q30min; PPH risk after birth
Calcium gluconate	Magnesium antidote
Labetalol, hydralazine, nifedipine	Antihypertensives; IV rescue for severe range
Methotrexate	Unruptured ectopic (and retained molar tissue); stop folic acid, no alcohol, serial hCG
RhoGAM	28 weeks; within 72 h if baby Rh+; any time blood may mix
Oxytocin	High-alert; own pump; LR main line; port closest to client
Dinoprostone, misoprostol	Prostaglandin ripening; remove for tachysystole/distress
Penicillin	GBS; q4h; ≥2 doses before birth
Anticoagulants (e.g., Lovenox)	Stop clot growth; watch for bleeding
Insulin	Doesn't cross placenta; needs rise in 2nd/3rd trimester